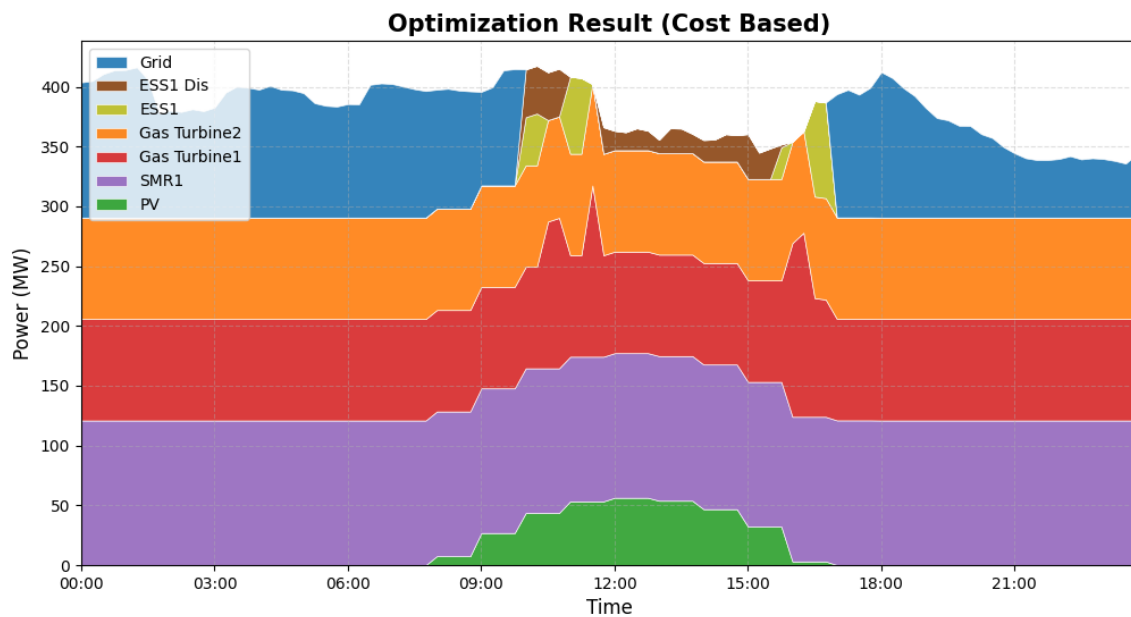


# <Energy Dispatch Report>



## 1. Microgrid Components

- PV: 30.0 MW
- GT: 100.0 MW
- SMR: 120.0 MW
- ESS: 40.0 MW / 160.0 MWh

## 2. Cost Optimization Result

- Baseload Generation Cost: 875 KRW
- Variable Cost: 4,000 KRW
- Total Cost: 5,000 KRW

## 1. Dispatch Optimization Analysis

- SMR: Baseload Generation
  - Due to its low generation cost, SMR operated as the baseload source with an average dispatch of 44.2 MW.
  - Provide a stable and economic supply across the full operating horizon.
- GT: Flexible Plants
  - Due to its higher generation cost, GT was mainly dispatched for flexibility and residual demand support at 19.2 MW on average.
  - Provide flexible response for load variation and high-price periods.
- PV: Peak Renewable Support
  - Due to its low generation cost, PV contributed 18.3 MW on average and 15.0 MWh in total.
  - Curtailment can be minimized when daytime demand is sufficiently high.
- ESS: Arbitrage and Peak Shaving
  - Reducing peak and providing flexibility with 7.5 MW average discharge and 6.2 MWh total discharge.
  - Through arbitrage and ancillary services, ESS supports time-shifting and operational resilience.

## 1. TOU Strategy

- Off-Peak
  - SMR: 42.5 MW
  - GT: 27.5 MW
  - PV: 5.0 MW
  - ESS: 2.5 MW / 67.5 MWh
- Mid-Peak
  - SMR: 50.0 MW
  - GT: 20.0 MW
  - PV: 20.0 MW
  - ESS: 0.0 MW / 80.0 MWh
- On-Peak
  - SMR: 40.0 MW
  - GT: 10.0 MW
  - PV: 30.0 MW
  - ESS: 20.0 MW / 60.0 MWh

## 2. Summary

- test

Detailed Simulation Data (24h)

| Time  | Grid | PV   | GT1  | SMR1 | ESS1 | Tot  | Dif   |
|-------|------|------|------|------|------|------|-------|
| 00:00 | 20.0 | 10.0 | 30.0 | 40.0 | 0.0  | 90.0 | -10.0 |
| 06:00 | 10.0 | 20.0 | 20.0 | 50.0 | 0.0  | 80.0 | -20.0 |

| Time  | Grid | PV   | GT1  | SMR1 | ESS1 | Tot  | Dif   |
|-------|------|------|------|------|------|------|-------|
| 12:00 | 0.0  | 30.0 | 10.0 | 40.0 | 20.0 | 70.0 | -30.0 |
| 18:00 | 15.0 | 0.0  | 25.0 | 45.0 | 5.0  | 90.0 | -10.0 |